EXPERIMENT 16

AIM:

To write and execute a Python program to implement a Feed Forward Neural Network (FFNN) for processing input data and predicting the corresponding output using forward propagation.

ALGORITHM:

1. Start the program.
2. Import the required libraries, such as NumPy and MLPClassifier.
3. Initialize the input dataset and corresponding output labels.
4. Create a feed forward neural network with an input layer, hidden layer, and output layer.
5. Train the network using the training dataset.
6. Predict the output for the given input data.
7. Display the predicted output and model accuracy.
8. Stop the program.

PROGRAM:

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```
import numpy as np
from sklearn.neural_network import MLPClassifier

X = np.array([
    [0, 0],
    [0, 1],
    [1, 0],
    [1, 1]
])

y = np.array([0, 1, 1, 0])

model = MLPClassifier(
    hidden_layer_sizes=(4,),
    activation='relu',
    solver='lbfgs',
    max_iter=2000,
    random_state=42
)

model.fit(X, y)

predictions = model.predict(X)

print("Input Data:")
print(X)

print("\nActual Output:")
print(y)

print("\nPredicted Output:")
print(predictions)
```

OUTPUT:



RESULT:

The Python program implementing a Feed Forward Neural Network was executed successfully. The network learned the AND logic operation and predicted the outputs for the given input combinations.